STANDARD OPERATING PROCEDURE (SOP) & DEVELOPER MANUAL

Wa CRM — External Proforma & Client Integration API

Document ID:	SOP-INT-WACRM-001	**Effective Date:**	October 2026
Production Base URL:	`http://www.wapluscrm.in`	**Version:**	1.0.0 (Production)
Target Endpoint:	`POST /api/integrations/proforma`	**Security:**	API Key (Header)

1. Connection Prerequisites & Authentication Credentials

All requests from your external website or backend must supply the authorized integration key via standard HTTP headers:

Parameter	Value	Notes
Base URL	`http://www.wapluscrm.in`	Production endpoint
Header Option A	`X-API-Key: wacrm_ext_live_8849204128f73a0e691823c91`	Recommended header
Header Option B	`Authorization: Bearer wacrm_ext_live_8849204128f73a0e691823c91`	Standard OAuth Bearer format
Rate Limit	**120 req / minute**	Sliding window per IP/key (HTTP 429 on breach)
Interactive Docs	`http://www.wapluscrm.in/docs`	Live Swagger sandbox to test endpoints

2. API Routes Specification

`POST`	`/api/integrations/proforma`	**Main Endpoint**: Ingests client & proforma, recalculates GST, enforces idempotency, returns authoritative totals.
`GET`	`/api/integrations/proforma/{id_or_number}`	Retrieves single authoritative proforma by numeric ID or Document Number (e.g. PI-2026-0042).
`GET`	`/api/integrations/proforma`	Lists and paginates authoritative proformas with search, customer ID, and limit filters.
`OPTIONS`	`/api/integrations/proforma`	CORS preflight handler for cross-origin web browser and client applications.

3. Duplicate Prevention & Authoritative Tax Calculation

Duplicate Prevention (Idempotency): Send a unique transaction UUID in header X-Idempotency-Key or body field idempotencyKey. If an identical request is repeated within 24 hours, the CRM re-returns the previously issued document (processingStatus: AUTHORITATIVE_VERIFIED) without creating duplicate records.
Authoritative Tax Rules: The CRM calculates Taxable Subtotal = (Rate * Qty) - Discount.
• **Intra-state (Tamil Nadu, State Code '33')**: 9% CGST + 9% SGST.
• **Inter-state (Other States / Union Territories)**: 18% IGST.
Arbitrary caller-supplied totals are verified; authoritative CRM math is enforced on the generated invoice.

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4. JSON Request Field Dictionary (`POST /api/integrations/proforma`)

Field	Type	Req?	Description & Rules
`idempotencyKey`	string	Yes*	Unique transaction UUID (prevents duplicate documents).
`companyName`	string	Yes**	Legal company or customer trade name.
`contactPerson`	string	Yes**	Primary contact name (**Either companyName or contactPerson required).
`phone`	string	Yes	10-digit phone number with optional country code.
`email`	string	Opt	Email address for billing and invoice dispatch.
`gstin`	string	Opt	15-character GSTIN (e.g. 33AAAAA1234A1Z5).
`billingAddress`	object	Opt	Object with line1, city, state, stateCode, pincode.
`proformaNumber`	string	Opt	Document number. If omitted, CRM auto-generates PI-YYYY-XXXX.
`productCategory`	string	Opt	wahrms (Software) or biometric (Hardware).
`lineItems`	array	Yes	List of purchased items (minimum 1 item required).
`item.name`	string	Yes	Product / Service bundle title.
`item.hsn`	string	Opt	HSN/SAC code (998314 for software, 85437099 for biometric).
`item.qty`	number	Yes	Quantity (must be > 0).
`item.rate`	number	Yes	Unit rate excluding tax (must be ≥ 0).
`item.discount`	number	Opt	Flat line discount (default: 0.0).
`item.gstRate`	number	Opt	Tax decimal (0.18) or percentage (18).

5. Live HTTP Request Example

```
POST /api/integrations/proforma HTTP/1.1
Host: www.wapluscrm.in
X-API-Key: wacrm_ext_live_8849204128f73a0e691823c91
X-Idempotency-Key: idem_req_20261017_001
Content-Type: application/json

{
"idempotencyKey": "idem_req_20261017_001",
"companyName": "Acme Global Solutions Pvt Ltd",
"contactPerson": "Rajesh Kumar",
"phone": "+91 9876543210",
"email": "billing@acmeglobal.com",
"gstin": "33AAAAA1234A1Z5",
"billingAddress": {
"line1": "Level 4, Tidel Park, Rajiv Gandhi Salai",
"city": "Chennai", "state": "Tamil Nadu", "stateCode": "33", "pincode": "600113"
},
"productCategory": "wahrms",
"lineItems": [
{
"name": "WAHRMS Core HR + Payroll Cloud Suite",
"hsn": "998314", "qty": 1, "rate": 75000.00, "gstRate": 0.18
}
],
"termsAndConditions": "1. Software onboarding commences upon payment receipt."
}
```

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6. Authoritative Success Response (HTTP 201 Created)

```
HTTP/1.1 201 Created
Content-Type: application/json

{
"success": true,
"requestId": "req_88a91b2c4d",
"processingStatus": "CREATED",
"customerId": 101,
"customerName": "Rajesh Kumar",
"companyName": "Acme Global Solutions Pvt Ltd",
"proformaId": 204,
"proformaNumber": "PI-2026-0042",
"authoritativeSummary": {
"taxableSubtotal": 75000.0,
"taxMode": "intra",
"cgst": 6750.0,
"sgst": 6750.0,
"igst": 0.0,
"totalTax": 13500.0,
"roundOff": 0.0,
"grandTotal": 88500.0,
"lineItemsCount": 1
},
"pdfDownloadUrl": "/api/integrations/waquote/proformas/204/download",
"syncedAt": "2026-10-17T11:10:00+00:00",
"message": "Proforma Invoice #PI-2026-0042 successfully registered in CRM."
}
```

7. Ready-to-Copy Developer Integration Code

Node.js (Axios):

```
const axios = require('axios');
const response = await axios.post('http://www.wapluscrm.in/api/integrations/proforma', {
idempotencyKey: `idem_${Date.now()}`,
companyName: 'Acme Corp', contactPerson: 'Ravi', phone: '+91 9876543210',
lineItems: [{ name: 'WAHRMS Software', qty: 1, rate: 50000, gstRate: 0.18 }]
}, {
headers: { 'X-API-Key': 'wacrm_ext_live_8849204128f73a0e691823c91', 'Content-Type': 'application/json' }
});
console.log('Invoice #:', response.data.proformaNumber, 'Grand Total:',
response.data.authoritativeSummary.grandTotal);
```

cURL (Terminal One-Liner):

```
curl -X POST "http://www.wapluscrm.in/api/integrations/proforma" \
-H "X-API-Key: wacrm_ext_live_8849204128f73a0e691823c91" \
-H "Content-Type: application/json" \
-d '{"companyName":"Test Client","contactPerson":"Ravi","phone":"9876543210","lineItems":[{"name":"WAHRMS","rate":5000
0,"qty":1,"gstRate":0.18}]}'
```

8. HTTP Status Codes & Troubleshooting

401	Missing server-to-server authentication...	Add header X-API-Key: wacrm_ext_live_8849204128f73a0e691823c91
403	Invalid API key or unauthorized credential	Verify API key characters. Check for trailing whitespace or typos.
422	Validation error (missing company/items)	Ensure companyName or contactPerson and lineItems are non-empty.
429	Rate limit exceeded (120 req/min)	Reduce call frequency or implement exponential backoff.

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